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Decentralization and Criminal Gangs in El Salvador: Impacts on Municipal Finances and Local Economic Development

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ABSTRACT *Recent literature has argued that decentralization can create new forms of leverage for criminal gangs, increasing their territorial presence and levels of violence. In this article we reverse the causal arrow and analyze how gangs affect the performance of decentralized institutions. We study the case of El Salvador, a country with the ubiquitous presence of gangs. We find that the higher presence of gangs reduced municipalities' fiscal revenues, thereby increasing their dependence on transfers from the central government. This result is mostly driven by mid-size municipalities, as the effect is not significant for small and large municipalities. In addition to depressing revenues, gang presence was also associated with lower municipal spending and less service provision, effectively hollowing out municipal governance. The drop in municipal spending is particularly strong in capital spending and in small municipalities. The article also finds that gangs undermine economic activity – not just in the municipalities where they operate but through spill-over effects in neighboring jurisdictions as well.*

KEYWORDS: gangs; decentralization; municipal finance; El Salvador

Decentralization helped El Salvador overcome the legacies of its deadly civil war, a twelve-year conflict (1980-1992) during which armed rebels unsuccessfully sought to overthrow a national government dominated by economic elites. While these elites continued to control the government in the years following the conflict through the *Alianza Republicana Nacionalista* (ARENA) party, the rebels created their own party that was able to win elections in a growing number of municipalities, including eventually in the capital city of San Salvador (Holland, 2016). Electoral rules that guaranteed a majority for the mayor's party meant that those municipalities under the control of the FMLN (*Frente Farabundo Martí para la Liberación Nacional*) became meaningful spaces of political opposition vis-a-vis ARENA at the national level. Meanwhile, many local governments introduced innovative participatory mechanisms to broaden decision making beyond partisan politicians (Bland, 2011), and fiscal decentralization

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gave municipal governments a more significant share of revenue. The transformation of local governments into spaces of political contestation laid the foundations for the FMLN's eventual presidential victories in 2009 and 2014 along with the practice of ideological alternation in power. Although El Salvador's two-party system has since been transformed by the rise of Nayib Bukele's New Ideas party, decentralization nevertheless deserves much of the credit for the political advances that El Salvador made in its emergence from internal armed conflict.

More recently, however, El Salvador's experience with decentralized governance has been undermined by a non-state armed actor of a different sort: criminal gangs. Gangs (or *maras*) date back to the 1960s, but the problem worsened when the United States deported thousands of convicted criminals after the end of the civil war, including Salvadorans who had joined either the Mara Salvatrucha (MS-13) or Barrio 18 in Los Angeles (Sviatschi, 2022). Over time, and partially in response to the adoption of hardline '*mano dura*' policies by both ARENA and FMLN governments, gangs transformed from 'loose neighborhood-level federations that operated largely autonomously and lacked internal governing bureaucracies' into 'hierarchically organized structures with organizational bases inside the country's prisons' (Moncada, 2021). With as many as 60,000 members, gangs have operated in almost all of El Salvador's 262 municipalities, where weaknesses in the rule of law enabled them to engage in widespread extortion activities. In El Salvador, as Eduardo Moncada argues, criminal taxation became a 'regular, costly, and humiliating facet of everyday life' (2021, 96). More specifically, as we show in this paper, extortion by gangs has had a devastating impact on the ability of decentralization to deliver on its promise in El Salvador.

The relationship between decentralization and security has generated a sizable literature, though much of it focuses on how this reform might encourage insurgent groups to lay down their arms – a question that was more relevant to El Salvador in the 1990s than it is today. Less emphasis has been placed on the link between decentralization and criminal gangs as a different type of non-state armed actor. To the extent that scholars have addressed this connection, they have asked how gangs took advantage of the institutional architecture created by decentralization. For example, according to Viridiana Rios (2013), decentralization worsens criminal violence because different levels of government often fail to act cohesively as a single decision-making body, enabling criminal organizations to go unpunished. On a similar note, as Angelica Durán-Martinez (2018) argues, decentralization is likely to increase the fragmentation of the state security apparatus along with the visibility of violence by criminal organizations. This is because 'the provision of reliable protection to criminals depends on the ability of state actors to guarantee that no other state authority will enforce the law against criminals' (12).

In this paper, we reverse the causal arrow between decentralization and gangs by arguing that, while the former may indeed create new forms of leverage for gangs, it is also the case that gangs directly undermine the ability of decentralized institutions to function as intended. In El Salvador, the territorial presence of gangs limited local fiscal autonomy because the rents they extorted from local populations had a depressing effect on the ability of municipalities to collect their own revenues and citizens' ability to pay. Vulnerable populations that are extorted by gangs had fewer resources to comply with their local fiscal obligations. Especially in mid-size municipalities, we show that the greater presence of gangs as measured by homicide rates reduced local tax revenues as municipalities struggle to collect the taxes and fees they are assigned by law. In addition to depressing tax revenues, the presence of gangs was associated with less municipal spending, limited-service provision, and greater dependence on revenue transfers from the central government. Beyond these negative fiscal outcomes, gangs also inhibited factor mobility between municipal jurisdictions. Indeed, we show that through such activities as neighborhood checkpoints that limit labor flows and extorted payments on the transport of goods, gangs undermined economic activity – not just in the municipalities where they operate but through spill-over effects in neighboring jurisdictions as well.

Our argument proceeds as follows. In the next section we review the literature on decentralization in the context of non-state armed actors. Section two describes the design of decentralization in El Salvador in the period before President Bukele came to power in 2019. The third

section explores the fiscal performance of municipalities across a number of dimensions and assesses the impact of gangs on fiscal and economic outcomes, demonstrating that gangs have an especially corrosive effect on the fiscal performance of mid-size municipalities, along with negative effects on service provision and local economic activity. Section four concludes.

1. Theoretical expectations about decentralization and non-state armed actors

If decentralization reassigns authority and resources to different levels of government within the same state, what happens if the state in question faces significant non-state armed actors? In many countries of the global south, the state does not enjoy a monopoly on the legitimate use of force and instead its attempt to control the national territory is contested by a range of challengers who themselves are able to exert territorial control (Luna & Giraudy, 2017). Scholars have engaged in a robust debate on the merits of decentralization in these contexts. According to the more optimistic strain of literature, decentralization can address some of the grievances that have led non-state actors to initiate rebellion in the first place, including a lack of responsiveness on the part of national governments in far-away capitals, and the inability of subnational governments to tailor policies to fit the particular ethnic and/or cultural practices of subnational regions (Bland, 2006, Brinkerhoff, 2011). Other scholars have struck a more cautionary note in arguing that decentralization can backfire. According to Dawn Brancati (2006), decentralization is likely to foment secesion if it empowers regionally-oriented political parties that stand to gain electorally by further fanning the flames of grievance. In other cases, rebels can penetrate into subnational governments and use decentralized resources to engage in service delivery, thereby enhancing their legitimacy as they continue to try to overthrow the central state (Eaton, 2006).

Not all non-state armed actors seek to overthrow or secede from the state, however. Compared to insurgents or guerrillas, criminal organizations like gangs pose a very different set of challenges to the state. The concern is not that gangs will use decentralized resources to strengthen their drive to secede from the country or bolster their prospects of defeating the government militarily. In contrast to rebel groups, as Lessing argues, 'criminal groups' opposition is subtler' (2021, 855). Attempts to better understand this more subtle relationship between criminal organizations and the state have produced an important new literature in recent decades. Some scholars have argued that, considering the prominent role that criminal organizations play in providing goods and services in the territories they control, it is appropriate to consider them as operating a 'shadow state' or a 'parallel state' (Chipkin & Swilling, 2018, Van Klinken, 2003). Others have posited a more symbiotic relationship between criminal organizations and the state, conceptualizing state rule and gang rule as complements (Blattman, Duncan, Lessing, & Tobón, 2021, Lessing, 2021) and emphasizing the mutual benefits that states and gangs can derive from each other (Koivu, 2018). One major study of Medellín (Blattman et al., 2021) finds that gang leaders deliver governance services in the neighborhoods they control largely to protect drug rents: 'gangs keep order in order to reduce civilian demand for and collaboration with police (p. 1),' which can 'generate strategic complementarities between gang and state rule' (p. 2). As Lessing concludes, 'we can no longer truly understand state governance in isolation from criminal governance' (2021, 869).

While state policies of incarceration certainly facilitated the organizational transformations that made gangs more effective in El Salvador, the dynamic we analyze in this paper is not one of symbiosis between gangs and the state. We emphasize instead a much more zero-sum and parasitic relationship between state rule and gang rule in large part because, unlike gangs elsewhere, neither MS-13 nor Barrio 18 have depended upon drug trafficking as a revenue-generating activity. According to Blattman et al. (2021), the absence of a retail drug trade in El Salvador – in stark contrast to Medellín – obviates the need to foster order, and as a result 'gangs can afford to be repressive and illegitimate' (p. 5). El Salvador thus represents an important case for theory building vis-à-vis the literature on the relationship between gangs and the state; the lesser interest in providing order to protect drug profits has generated a far less symbiotic relationship with the

state. Just as importantly, as we show below, the lesser reliance on drug profits comes coupled with a very heavy reliance on extortion. Thus, while our findings might not be generalizable to cases where gangs rely more heavily on drug trafficking, there is nothing specific to El Salvador about the core argument; indeed where gangs in other contexts also rely especially heavily on extortion we would expect to see the same impacts that are documented in our analysis.

Our main contribution is to show that this reliance of gangs on extortion can disrupt the various causal mechanisms that have led theorists to posit a positive relationship between decentralization and improved developmental outcomes.[2] One of the main arguments for decentralization is that it enables subnational governments to compete with one another through factor mobility (Oates, 1972, 2005; Tiebout, 1961). Labor and capital may more easily flow between neighboring subnational jurisdictions than they can between countries, and voters can vote with their feet by opting to relocate in local jurisdictions whose policy offerings better match their preferences (Ter-Minassian, 1997). Competition between subnational governments for capital investment and human resources encourages local politicians to adopt 'market-preserving' policies that enhance prospects for economic growth (Weingast, 1995).

Through a number of illicit behaviors, however, gangs can limit factor mobility by extracting payoffs from economic actors seeking to transport goods between municipalities, and by imposing checkpoints that limit the mobility of local residents.[3] Criminal organizations interact with businesses in ways that powerfully affect the local economic environment in which they operate (Fenizia, 2018), not just in illegal markets but in formal markets as well, where they can affect levels of production, employment, and prices (Ferrante, Fontana, & Reito, 2021; Montoya, 2016; Rozo, 2014). In some cases, scholars have documented salary variations that are due chiefly to restrictions imposed on the mobility of less skilled labor in localities with a high gang presence; higher labor costs due to these restrictions may result in higher production costs (Rozo, 2014). Widespread extortion of local businesses by gangs prohibits them from playing roles as partners in municipal-level security reform (Moncada, 2016).

Gangs can also disrupt another important mechanism through which decentralization is intended to improve fiscal outcomes, which is the ability of local governments to collect own-source revenues. When decentralization takes the form of devolving tax bases to subnational governments, it can have a host of positive impacts for accountability, encouraging local tax payers to demand that their local governments use own-source revenues to provide services (Azfar, Kahkonen, Lanyi, Meagher, & Rutherford, 2004). At the same time, the political and administrative costs associated with collecting unpopular local taxes can lead to anemic local tax effort, discouraging local authorities from collecting the tax bases they have been assigned. While insufficient local tax effort deserves the attention it has received from scholars of decentralization (Faguet, 2014), it is important to note that widespread gang activity may also be impeding efforts by local authorities to collect taxes. Impoverished residents may be unable to pay 'rent' to the criminal gangs who are directly and violently extorting them, while at the same time complying with the municipal taxes and fees they are legally obligated to pay. If gangs and mayors are trying to tap the same resources, there are good reasons to expect the former will win out. The negative impact of gangs on the ability of municipal governments to collect taxes and fees operates as a further impediment to the general norm that 'finance should follow function' (Rodríguez-Pose & Vidal-Bover, 2022).

When gangs inhibit the collection of own-source revenues, they can trigger a number of other negative fiscal outcomes, including municipal dependence on fiscal transfers from the center (Rodden, Rodden, Eskeland, & Litvack, 2003). A large literature has emphasized the perverse incentives that transfer systems can generate, especially when they take the form of non-earmarked revenues (Bahl & Martinez-Vazquez, 2006; Inman & Rubinfeld, 1997). Transfer systems can also increase the scope for politically-oriented discretion on the part of the national government. The inability to collect local revenues because criminal gangs are tapping the same resource base may also lead to increased levels of local debt and pressure for bailouts (Di Cataldo & Mastrorocco, 2021).

If criminal gangs can obstruct local tax collection, increase the reliance on transfers, and worsen local debt, they can also powerfully affect the relationship between decentralization and service provision by local governments. Here scholars have pointed to two problematic dynamics. The first is that the presence of gangs may make it more difficult for neighboring municipalities to use rival approaches to service provision as a means of differentiating themselves. Di Cataldo & Mastrorocco (2021) argue that municipalities captured by organized crime experience a distortion in the provision of public services due to a variety of rent seeking activities. Bueno (2022) and Cordova (2017) likewise document significant gang impacts on service provision. In addition to shaping the mix of service provision, gangs can impact the overall level as well. Several scholars have also emphasized how organized crime reduces the incentives and capacity of local governments to provide critical goods and services across the board, starting with the protection of citizens (Acemoglu, De Feo, & De Luca, 2020; Arends, 2021; Beraldo, Bordignon, Pellegrino, Piacenza, & Turati, 2017; Melnikov, Schmidt-Padilla, & Sviatschi, 2020; Pinotti, 2015).

Through the multiple mechanisms examined in this section, gangs have not only increased the fragility of cities across El Salvador (Nogueira, 2017), but indirectly prevented decentralization from generating the positive outcomes that it is adopted to deliver.

2. The evolution of decentralization and local finances

This section describes two distinct but intersecting territorial dynamics in El Salvador: the design of measures that decentralized resources and authority from the national government and the impact of those measures on local fiscal performance.

Reflecting its unitary status and highly centralized past, El Salvador maintains just one level of subnational government composed of 262 municipalities that vary significantly in population, economic dynamism, and levels of urbanization. At the intermediate level, El Salvador is organized into 14 departments whose function is to represent the central state and whose authorities (governors) are appointed by the national government rather than elected in their own right. Although decentralization occurred just to the municipal level, it affected all three dimensions of change: political, administrative, and fiscal.[4]

In the political sphere, electoral rules for municipal elections allow parties to present closed lists that are topped by the mayoral candidate and followed by as many candidates as there are seats on the council (Córdova & España-Nájera, 2021). Prior to the 2015 electoral reform, the winning party was guaranteed complete control of the municipal council with no opposition seats. Since then, opposition parties are allowed to gain some representation on the council while still guaranteeing a majority for the mayor's party. Despite this highly majoritarian design, political decentralization led to the emergence of bi-partisan coordination among mayors within the Council of Municipalities of the Republic of El Salvador (COMURES), which sought to defend the roles played by municipalities vis-à-vis the center (Bland, 2011). In addition to elections, the 1980 municipal code in El Salvador provides for a number of other mechanisms to enhance participation, including local development councils and open assemblies several times a year. As Bland argues, 'local government allowed for alternation in executive power and demonstrated that turnover of government from one party to another was possible without inciting violence' (Bland, 2011, 866).

El Salvador is unusual in Latin America in that it denies municipalities the ability to levy property taxes; municipal governments instead can collect direct fee-based taxes paid by individuals and companies that develop economic activities within the district. Municipalities can also raise non-tax revenues from user fees and charges, as well as revenue from the sale of goods and services. In addition to those own-source revenues, municipalities received transfers through a revenue sharing system that was developed beginning in 1988 as the Fund for Municipal Economic and Social Development (FODES). FODES funds are distributed

according to population, equity, poverty, and land area criteria, and the municipalities that receive them are required to devote 80% of the revenues to capital investment projects and no more than 20% on operating expenses.

During the period 2010-2018, municipal finances in El Salvador were characterized by two key features: (i) high reliance on intergovernmental transfers, with the annual average of FODES representing 48% of total municipal revenues, and (ii) low generation of own-source revenues, which represented about 37% of total municipal revenues. For small and medium-size municipalities, FODES represented an especially significant share of total revenues. In 2018, FODES transfers were 10 times the average own-source revenues for small municipalities (less than 20,000 inhabitants) and 3.5 times the average for medium-size municipalities (between 20,00 and 45,000 inhabitants). For larger municipalities, the FODES/own-source revenue ratio was 1.2 (see Table 1).

Own-source revenues represented 38.8% of total municipal revenues in 2018. The greater the population size of a municipality, the greater the revenues collected (see Figure 1). For example, whereas Antiguo Cuscatlán (La Libertad) collected $508.5 of own-source revenues per capita, Nuevo Cuscatlán collected $393, and San Salvador collected $251, smaller, more rural municipalities collected only around $4 per capita in own-source revenues.

In aggregate, for 2018 the average expenditure[5] of municipalities in per capita terms was $159, with only 32% of local expenditures financed by own-source revenues. However, differences in size and economic activity are critical: the greater the population size, the greater the share of own-source revenues that finance municipal expenditures. Figure 2 below shows that municipalities in the most populated and developed departments of El Salvador, namely, San Miguel, Santa Ana, La Libertad, and San Salvador, are less dependent on FODES.

3. Fiscal and economic impacts of gang violence at the local level

To evaluate the effect of gang presence on local finance, a panel data model with fixed effects was constructed using data from the 258 municipalities in El Salvador for the years 2010-2019. We evaluated the effect of gang activities on fiscal outcomes, and on the quality of public service provision, using information about solid waste collection (measured as the percentage of the district population that disposes of waste through public collection).

We operationalized the presence and intensity of gangs using homicide rates as a proxy (the variable that we have labelled Mara Salvatrucha) as well as other control variables containing specific characteristics of each municipal jurisdiction. Additionally, we considered complementary specifications including the number of homicides attributed to gangs other than MS-13

Table 1. Per capita tax collection and size of FODES transfers, 2011–2018

	Small		Medium-sized		Large	
Year	Tax collection PC	Ratio of FODES own revenues	Tax collection PC	Ratio of FODES own revenues	Tax collection PC	Ratio of FODES own revenues
2011	13.8	25.0	27.7	4.2	35.1	1.2
2012	13.9	13.7	29.0	5.1	36.5	1.4
2013	15.7	12.4	31.9	4.7	39.0	1.3
2014	16.6	13.5	31.1	4.5	41.4	1.2
2015	17.1	14.5	33.0	4.5	43.9	1.2
2016	18.4	13.0	35.0	4.2	46.3	1.2
2017	19.2	11.1	34.2	3.8	46.2	1.1
2018	21.5	10.0	39.0	3.5	51.5	1.2

Source: Authors' elaboration.

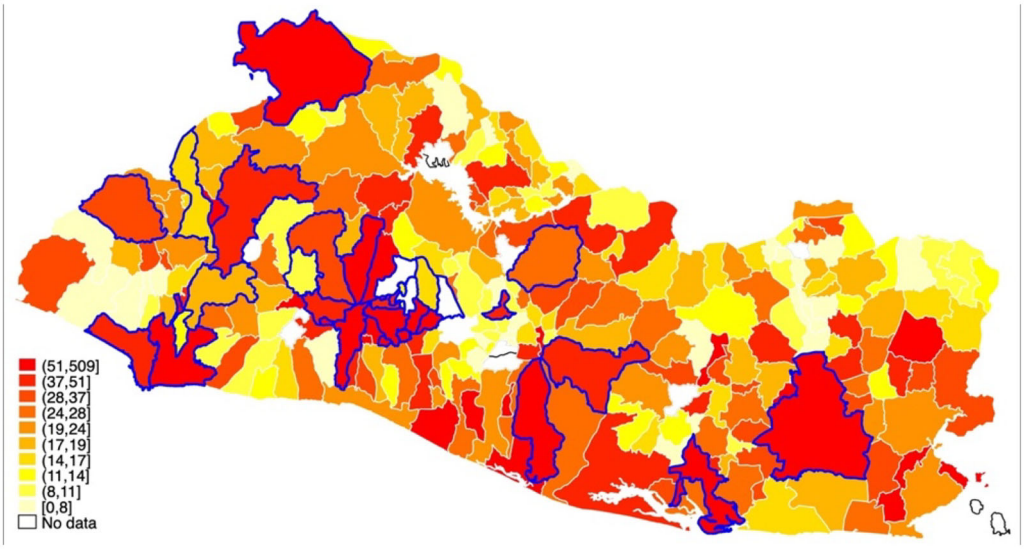

Figure 1. Own revenues, per capita - 2018.
*Municipalities border in blue are those with more than 50,000 inhabitants.
Source: Portal de Transparencia Fiscal - El Salvador; ISDEM.
Authors' elaboration.

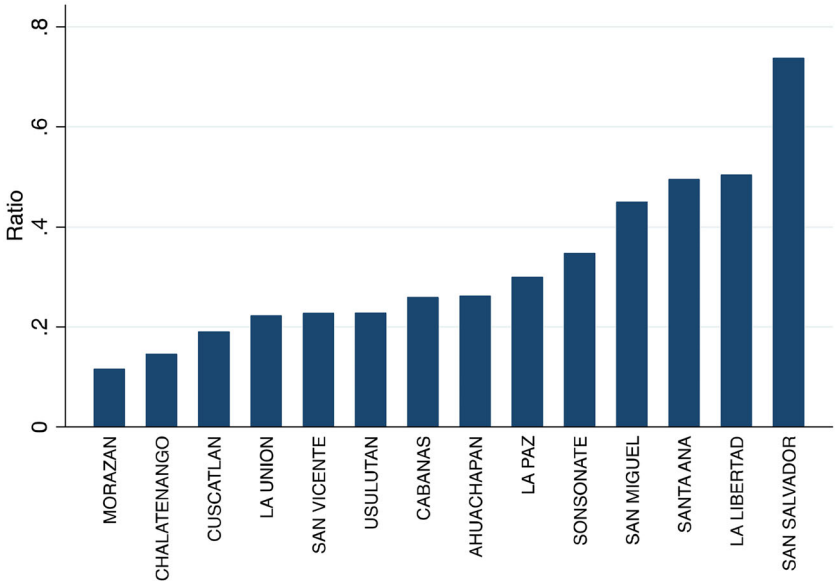

Figure 2. Budgetary autonomy in 2018.
Municipalities Aggregated by Department.
Source: Portal de Transparencia Fiscal - El Salvador; ISDEM.
Authors' elaboration.

(see Supplementary Materials, Figure 1 and Tables 1 and 2). The idea is to identify differences in the presence of different types of gangs, which would indicate that gangs with a high level of organization and territorial presence, as was the case with MS-13, may generate distinct responses compared to unorganized gangs that operate in random places.

We argue that gangs' illegal activity in the territory introduced a non-formal competing authority due to their credibility to collect and enforce illegal revenue collections, in the form of

Table 2. Main statistics of variable population by municipality - 2019

Statistics		Percentiles	
Mean	22,238	10%	3,158
Std. dev.	38,047	25%	5,486
Variance	1.45E + 09	50%	10,076
Skewness	5	75%	21,675
Kurtosis	28	90%	52,359
Obs	245	99%	241,403

Table 3. Effect of MS-13 violence on per capita own-source revenue

	(1) All municipalities	(2) Population < 10k	(3) 10k < Population < 50k	(4) Population > 50k
Maras	−1.136*	−0.426	−5.613*	−22.287
	(0.829)	(0.623)	(2.864)	(17.779)
Kw_mm	0.217	0.133*	0.853*	0.148**
	(0.125)	(0.179)	(0.435)	(0.054)
UrbanPopul	−0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Transf_pc	−0.130*	−0.015	−0.095	0.740
	(0.049)	(0.012)	(0.092)	(0.627)
Density	−0.016	−0.056	0.245	6.105
	(0.129)	(0.161)	(0.982)	(7.532)
Garbage_sv	2.204	0.451	10.871***	17.144
	(1.971)	(1.341)	(3.853)	(19.793)
year = 2012	2.151	−1.163	1.669	−3.070
	(1.277)	(2.761)	(1.841)	(3.724)
year = 2013	3.877*	−0.972	1.200	0.012
	(1.563)	(2.986)	(1.455)	(4.234)
year = 2014	6.029***	0.103	4.486	−1.869
	(1.800)	(3.136)	(2.769)	(5.912)
year = 2015	8.008***	1.394	6.095	0.774
	(1.979)	(3.065)	(3.203)	(6.177)
year = 2016	10.145***	1.122	9.722*	3.307
	(2.398)	(3.072)	(3.891)	(7.078)
year = 2017	10.681***	2.007	9.856**	4.288
	(2.202)	(3.014)	(3.568)	(6.013)
year = 2018	13.265***	3.964	10.250**	9.977
	(2.655)	(3.241)	(3.588)	(8.303)
Constant	27.078***	15.305***	9.079	−5.850
	(5.487)	(3.172)	(7.049)	(26.326)
Observations	1308	465	646	197

Standard errors in parentheses.
Robust standard errors in parentheses.
*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

extortion. This competing non-formal authority eroded a municipality's authority and capacity to collect revenues in the areas of a municipal jurisdiction where gangs exerted territorial authority. We find that when homicides attributed to gangs increase in a municipality, there was a significant drop in a municipality's own-source revenues per capita (see Table 3 below).

To analyze the impact of gangs in municipalities with different size of populations, the literature identifies a direct and positive relationship between city size and the incidence of criminal activities (Gaigné & Zenou, 2015; Glaeser & Sacerdote, 1999; Kahn, 2010). We identified a non-linearity in the relationship between gangs and local finances, depending on population

size. The non-linearity can be explained by economies of scale and increased control in tax collection as the size of the tax base increases (Canavire-Bacarreza & Espinoza, 2010). We created a model that includes specifications that capture the wide-ranging differences in population across municipalities. The hypothesis behind this is that larger municipalities (area size and population) have a greater level of financial autonomy and more diverse economic structure, and therefore greater capacity to secure funding for the provision of public services. Thus, gangs have a lesser impact on collection capacity for large municipalities. On the other hand, medium-sized municipalities, which are more dependent on FODES transfers, have a weaker capacity to generate their own revenues, making their finances and capacity to provide services more vulnerable to the gang. In the case of smaller municipalities, which have very limited capacity to generate their own revenues and heavily rely on transfers, the effect of gangs on revenue collection could be much smaller.

For the purpose of the model, municipalities were categorized as follows (as shown in Table 2): (i) municipalities with a population of less than 10,000 inhabitants, a value derived from the series' median; (ii) municipalities with a population between 10,000 and 50,000 inhabitants, and (iii) municipalities with a population greater than 50,000 inhabitants.

The following specification is therefore applied in each case:

$$Y_{i,t} = \beta_0 + \beta_1 MaraSalva_{i,t} + \beta_k X_{i,t} + \sum_{t=1}^{T-1} \gamma_t D_t + a_i + \mu_{i,t}$$

Where $Y_{i,t}$ is the dependent variable that will be, according to each model, the variables: (i) own source revenue per capita, (ii) expenditure per capita, and (iii) provision of public services. The variable $MaraSalva_{i,t}$ represents the presence of gangs as measured by the number of homicides per thousand inhabitants attributed to MS-13 per municipality. The number of homicides per thousand inhabitants allows us to measure not only the presence of gangs, but the intensity of criminal activities. In all models this variable is lagged by one period to help mitigate concerns related to simultaneity and reverse causality between the presence of gangs and fiscal outcomes.

The term β_0 denotes the intercept term, which captures the average value of the dependent variable (own revenues, expenditure, and debt in our different specifications) across all municipalities and across time. Term β_1 is the coefficient of the variable of interest (Maras, or in complementary specifications the variable that captures the presence of other gangs). Also included is a vector with other control variables $X_{i,t}$ such as percentage of urban population, population density, average households' income, average energy consumption, investment in capacity building within local governments, among others. Additionally, a vector is included with time dummy variables representing different time periods and a vector a_i, representing the individual-specific random effect. Random effects are included under the assumption that individual entities have unique characteristics that remain constant over time and may influence the relationship between the dependent variable and other independent variables. In this context, while examining fiscal outcomes and the presence of gangs, random effects could encapsulate unobservable factors such as local governance dynamics, economic conditions, social and cultural norms, historical events, institutional variations, and demographic characteristics unique to each jurisdiction. Therefore, the analysis of the coefficient related to gangs (β_1) focuses on understanding how the presence of gangs impacts revenues while considering underlying regional heterogeneity.

To identify the effects of gangs on local economic activity, as well as to measure whether gang activities in a municipality have spillover effects in adjacent municipalities, we developed a model that adds local economic activity as a dependent variable. Due to the absence of local information such as municipal GDP, we use the level of energy consumption (in kilowatt-hour) in the locality as a proxy of economic activity.

To measure the impact of gang activities at the municipal level and the uncertainty that gangs generate, a cross sectional spatial autoregressive model[6] is included, considering an adjustment estimator using a generalized spatial two-stages least squares *(GS2LS estimator)* following the procedure developed by Kelejian & Prucha (1998). The specification is as the following:

$$Y_i = \beta Gangs_i + \gamma Z_i + \rho W_{y;i} + \tau W_{X;i} + \varepsilon_i \tag{1}$$

$$\varepsilon_i = \lambda W_{\varepsilon,i} + \mu_i \tag{2}$$

Where:

- $Y_{i,t}$ is the dependent variable that seeks to include the level of energy consumption as a proxy variable.
- The variable $Gangs_i$ represents the presence of gangs using the homicides from Mara Salvatrucha (measured as the number of homicides per thousand inhabitants). To avoid the possibility of double causality, a lagged-value of the variable was considered.
- Additionally, $Z_{i,t}$ is a vector that includes other control variables, among which are the level of intensity of street night lighting[7], the urban population, family incomes, per capita expenditure and the percentage of the population aged over 65.
- The matrix 'W' represents a 'spatial weights matrix', which contains the spatial arrangements of the different localities to be analyzed and each entry represents the spatial weight associated with locality 'i' and locality 'j'. In this case if 2 municipalities share a border, the value is equal to 1, and zero otherwise. Due to this specification, the matrix allows the spatial measurement of different variables. The specification includes three types of interactions that incorporate a spatial component captured by the W matrix, that is: (i) an endogenous interaction, which represents the effect of the dependent variable (economic activity) on the economic activity of other provinces, captured by the estimator $'\rho'$ which measures the effect on the economic activity (Kwh consumption) in neighboring municipalities; (ii) the exogenous effect captured by the estimator $'\tau'$ of the variable $W_{X;i}$ which measures the effect of gangs' presence on the economic activity of neighboring localities, and (iii) the last interaction, included in the 'error' vector (λ), which captures other unobserved variables with some degree of precision. W_ε for other unobserved variables with some degree of spatial structure in neighboring localities.

It is important to note the strong heterogeneity that exists among different municipalities. Regarding the level of own-source revenues per capita for 2018, there is an average collection of \$28.9; however, the municipality with the lowest collection records an average of only \$2.42 per capita, compared to \$508 per capita in the municipality with the highest collection. In the case of expenditure per capita, the values also have a wide range, with an average per capita expenditure of \$143. The series maintains a per capita expenditure range from \$30.9 to \$546 per capita in the same year.

For the model associated with the impact of gangs on the quality of public services, the maximum theoretical level of service is 100% of all households in the municipality, and we find that a typical garbage collection service covers 48% of all households. In regards to the model that measures a municipality's consumption of electricity (both private and public), there is an average consumption of 20.58 million Kwh, while the municipality with the highest consumption in 2018 records 986 million Kwh compared to an average consumption of 0.2 Kwh in the minimum value. Please see Appendix for information sources as well as a discussion of endogeneity issues.

3.1. Results

3.1.1. Effects of gangs on local finance. Results obtained from the specifications of fiscal outcomes at the local level show that gangs have had an effect on the level of revenue collection, particularly in medium-sized municipalities, as well as effects on current expenditure. This reflects a level of vulnerability for a significant group of municipalities that have been affected by gang presence, both in their ability to sustain themselves through own-source revenues and to maintain the functioning of the entity and the provision of goods and services. The results are similar when including gangs other than MS-13 in the regressions.

In the specification using own-source revenues as the dependent variable and violence by MS-13 as the main independent variable, we found a high sensitivity in revenue collection; especially in medium-size localities. The different impacts according to different models that separate the samples of municipalities based on population size support the hypothesis of nonlinearity in the effects.

As seen in Table 3, column (2) shows that small municipalities do not have a statistically significant impact on own-source revenues due to gang activity. These municipalities had low per capita own-source revenues and were highly dependent on FODES transfers. Thus, their financial flow seems to be less influenced by violence within their jurisdiction. About 130 municipalities have less than 10,000 inhabitants, and within this group, the average per capita collection is US$32; indicating that the majority of them are below the average. Additionally, within the same group, the average per capita transfers amount to US$170 for the same year (2018). In the case of medium-sized municipalities, shown in column (3), the presence of MS-13 has a negative and significant impact on own-source revenues. Medium-sized municipalities typically have a more fragile and less diversified economy comprised of small and medium-sized businesses that are more vulnerable to the local social and economic context. Additionally, medium-sized municipalities may have limited resources to contain and reduce gang activities. In large municipalities, which host a diverse group of industries and commercial activities, there are no statistically significant impacts of gang activity on per capita own revenues. According to the data presented in the Supplementary Materials (Table 3), the presence of gangs other than MS-13 has no effect on local per capita revenue.

We found that the presence of MS-13 has a significant and negative impact on municipal spending per capita (see Table 4). Specifically, for each 1% increase in homicides by gangs, the per capita expenditure is expected to decrease by US$17.9 dollars. The greatest impact is on the smallest municipalities, where the drop in expenditures per capita for every homicide per 1000 inhabitants is US$20. When we included Other Gangs as an interest variable, the results are quite similar, showing the negative effect of gang presence on local expenditure, particularly in small municipalities (see Supplementary Materials, Table 4).

For robustness, additional estimations were included to determine the MS-13 effect on current expenditure (which includes expenditure on goods and services and on personnel and social obligations) and capital expenditure (local public investment) in per capita terms. These specifications are presented in Table 5. As seen in overall results, the impact on current and capital expenditure is significative and negative. When we disaggregated current and capital expenditures, we found that for every extra homicide per 1,000 inhabitants, capital expenditures drop by US$19.7, and current expenditures drop US$2.9.

In the case of capital expenditure, the fear of gang reprisals can also hinder the smooth execution of public works, affecting project timelines and escalating costs. These results may suggest a lack of territorial control, which is consistent with the literature characterizing the social and institutional effects of gangs, where gang domination of territory limits the entry of people or personnel not recognized or approved by the gangs who control the territory. This situation affects municipal and non-municipal employees whose work is critical for service provision. Cordova (2017) associates gang-controlled territory with higher levels of poverty and lower levels of municipal services. Additionally, the size of the locality seems to be a relevant factor in

Table 4. Effect of MS-13 on expenditures

	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k
Maras	−17.694***	−20.112**	−0.489	−38.447
	(6.347)	(12.785)	(7.180)	(29.544)
Kw	−0.006	−1.788*	0.137	−0.033
	(0.084)	(1.590)	(0.00)	(0.020)
UrbanPopul	−0.000***	0.001	−0.000	−0.000**
	(0.000)	(0.001)	(0.000)	(0.000)
Total_revenue_pc	1.077***	2.454***	0.891***	0.675***
	(0.210)	(0.611)	(0.218)	(0.184)
Density	8.661*	7.798*	−4.102	−24.177
	(2.687)	(2.425)	(2.848)	(16.557)
year = 2012	−6.980	−10.246	−6.819	6.558
	(3.843)	(9.701)	(3.643)	(3.620)
year = 2013	−5.096	−10.290	−2.824	5.885
	(4.470)	(10.244)	(4.646)	(4.177)
year = 2014	15.228**	18.975	12.621**	11.848
	(4.676)	(12.534)	(4.333)	(7.225)
year = 2015	−1.229	−12.911	0.766	1.515
	(3.937)	(10.887)	(4.687)	(3.756)
year = 2016	−14.906***	−37.984***	−8.827	−0.054
	(3.937)	(11.006)	(4.741)	(6.630)
year = 2017	−2.615	−20.951	2.251	1.650
	(4.001)	(13.647)	(4.994)	(5.471)
year = 2018	−14.213**	−46.045***	−2.254	−7.461
	(4.597)	(12.693)	(5.082)	(6.557)
Constant	85.488***	50.137	84.611***	35.412***
	(9.396)	(29.655)	(5.641)	(4.288)
Observations	1331	479	655	197

Standard errors in parentheses.
Robust standard errors in parentheses.
*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

these effects. In different specifications according to number of inhabitants, localities with a population less than 10,000 reflect a statistically significant coefficient while in larger municipalities there are no significant impacts on current or capital expenditure (see Supplementary Materials, Table 6).

These findings highlight the importance of considering the size of the locality when studying the impact of MS-13 activities on different types of expenditures. The results suggest that small-sized localities are particularly affected by the presence of MS-13, leading to significant reductions of expenditure. Further investigation may be necessary to explore the underlying factors contributing to these varying effects across different population sizes.

3.1.2. Effects of gangs on public service provision. We found a significant and negative correlation between the presence of MS-13 and service provision, particularly in medium size governments.[8] Results show that for each homicide per 1,000 inhabitants, the coverage of public services (garbage collection) drops by 0.2%. The model does not show any impact on public service provision for small and large municipalities (see Table 6).

Complementary estimation was conducted considering the number of homicides associated with other gangs and the results are similar in terms of the impact in medium-size municipalities (see Supplementary Materials, Table 5).

Table 5. Effect of MS-13 on expenditures according to size

	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k
Maras	−2.866**	−3.884**	2.426	−15.151	−19.702***	−16.949*	−5.381	−9.386
	(1.058)	(1.193)	(2.328)	(13.248)	(4.639)	(7.578)	(7.044)	(26.668)
Kw	0.076***	−0.587	0.133***	0.008	−0.060	−1.388	−0.111	−0.028
	(0.017)	(0.324)	(0.036)	(0.013)	(0.058)	(0.940)	(0.088)	(0.024)
Urban_popul	−0.000***	−0.000	0.000	−0.000	−0.000***	0.001	−0.001***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)
Transf_pc	0.557***	0.367***	0.526***	0.764***	0.352***	1.884***	0.133	0.429***
	(0.025)	(0.041)	(0.040)	(0.036)	(0.072)	(0.189)	(0.068)	(0.071)
Density	0.625**	0.580*	−0.315	−6.328	6.150***	7.117***	−2.580	−13.676
	(0.230)	(0.237)	(0.829)	(5.975)	(1.032)	(1.470)	(2.560)	(12.242)
year = 2012	3.472***	3.617**	3.831***	2.436	−9.808**	−13.757	−10.330**	4.255
	(0.840)	(1.304)	(1.151)	(2.316)	(3.686)	(8.657)	(3.649)	(4.766)
year = 2013	6.006***	6.342***	6.253***	3.705	−9.851**	−16.397	−8.229*	2.579
	(0.865)	(1.382)	(1.180)	(2.328)	(3.778)	(8.988)	(3.672)	(4.790)
year = 2014	7.682***	10.061***	7.777***	3.264	9.239*	9.893	6.216	9.025
	(0.874)	(1.394)	(1.195)	(2.333)	(3.789)	(9.021)	(3.685)	(4.797)
year = 2015	7.099***	10.290***	7.174***	0.971	−6.083	−21.976*	−4.612	1.138
	(0.885)	(1.400)	(1.241)	(2.418)	(3.797)	(8.948)	(3.736)	(4.957)
year = 2016	5.153***	7.172***	4.629***	3.903	−16.436***	−43.127***	−10.737**	−3.314
	(0.933)	(1.478)	(1.363)	(2.680)	(3.958)	(9.303)	(3.958)	(5.462)
year = 2017	7.660***	9.818***	6.843***	6.867*	−6.916	−27.692**	−1.818	−4.680
	(0.976)	(1.618)	(1.418)	(2.799)	(4.137)	(10.065)	(4.101)	(5.696)
year = 2018	7.554***	9.008***	8.392***	2.947	−17.908***	−51.450***	−7.456	−9.421
	(1.010)	(1.679)	(1.391)	(2.683)	(4.218)	(10.409)	(3.938)	(5.464)
Constant	23.018***	35.489***	19.515***	13.220***	70.934***	25.025*	73.683***	24.461***
	(1.604)	(2.921)	(2.808)	(3.021)	(4.949)	(11.479)	(4.093)	(5.795)
sigma_u								
Observations	1331	479	655	197	1331	479	655	197

Standard errors in parentheses.
Robust standard errors in parentheses.
*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

Table 6. Effect of MS-13 on public service provision

	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k
Maras	−0.002*	0.004	−0.002*	0.000
	(0.001)	(0.006)	(0.001)	(0.000)
Urban_popul	0.000***	0.000	0.000**	0.000*
	(0.000)	(0.000)	(0.000)	(0.000)
Household_income	0.000***	0.000**	0.000***	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)
PC_own_rev	0.000	−0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.001)
Density	0.001	0.000	0.002	−0.012
		(0.006)	(0.010)	
	(0.006)			(0.028)
MunCapacBuild	0.025	0.049*	−0.010	−0.032
	(0.015)	(0.023)	(0.020)	(0.020)
year = 2011	−0.014	−0.031*	−0.006	0.011
	(0.010)	(0.016)	(0.015)	(0.009)
year = 2012	0.002	0.023	−0.018	0.009
	(0.011)	(0.021)	(0.013)	(0.009)
year = 2013	0.006	0.029	−0.009	0.016
	(0.018)	(0.038)	(0.021)	(0.016)
year = 2014	0.016	0.029	0.006	0.026*
	(0.019)	(0.041)	(0.023)	(0.012)
year = 2015	0.024	0.046	0.004	0.020
	(0.019)	(0.038)	(0.023)	(0.013)
year = 2016	0.041*	0.089*	0.012	0.007
	(0.020)	(0.041)	(0.024)	(0.018)
year = 2017	0.025	0.072	−0.005	0.008
	(0.021)	(0.045)	(0.026)	(0.015)
year = 2018	0.034	0.086	0.006	0.017
	(0.023)	(0.047)	(0.026)	(0.014)
Constant	0.194***	0.205**	0.085*	0.517***
	(0.042)	(0.070)	(0.037)	(0.059)
Observations	1391	520	662	209

Standard errors in parentheses.
Robust standard errors in parentheses.
*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

3.1.3. Effects of gangs on local economic dynamics. We find a significant negative effect of gang extortion on local economic activity and spillover effects to neighboring localities. Results indicate that for every homicide that occurs in one locality attributed to MS-13 and to gangs in general (results for the last specification is shown in Table 6 in the Supplementary Materials), the consumption of electricity within the local territory is reduced by 3.1 million kWh and consumption in neighboring localities is reduced by 5.2 million kWh (see Table 7 below).[9]

4. Discussion

In El Salvador, the territorial control of gangs is clearly associated with lower municipal tax revenues, lower municipal expenditures, and decreased municipal capacity to provide public services. We found a high sensitivity in revenue collection; an increase in MS-13 homicides per 1,000 inhabitants leads to a reduction in per capita revenue of US$1.1 dollar. The effect is particularly relevant in medium-size localities, where an increase in one homicide per 1,000 inhabitants by MS-13 reduces per capita collection by US$5.6 dollars. We also found that the

Table 7. Impact of MS-13 on economic activity

	kWh
MaraSalva	−3.107**
	(0.996)
luminosity	0.013***
	(0.001)
PobUrb	0.000
	(0.000)
ingfa	0.059**
	(0.020)
gasto_pc	−0.005
	(0.011)
agedpopulation	143861
	(87197.9)
Constant	−51.824***
	(13.071)
Wma	
MaraSalva	−5.250*
	(2.149)
kw_mm	0.074
	(0.081)
All_gangs	
Observations	144

Standard errors in parentheses.
Robust standard errors in parentheses.
*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

presence of MS-13 has a significant and negative impact on local and illegal activities that lower municipal spending per capita. Specifically, for each 1% increase in homicides by gangs, the per capita expenditure is expected to decrease by US$17.9 dollars. Here the greatest impact is on the smallest municipalities, where the drop in expenditures per capita is US$20 for every homicide per 1000 inhabitants. Adverse impacts on own-source revenues and levels of local expenditure within affected municipalities can be attributed to the gangs' ability to control territorial areas, allowing them to impose taxes on the movement of factors (including people, public transport, and goods and services) between municipal jurisdictions, a result that was also documented in Brown, Montero, Schmidt-Padilla and Sviatschi (2020).

In terms of public service provision, there is a significant and negative correlation between the presence of gangs and service provision, particularly in medium size governments, where garbage collection decreased by 0.2% per 1,000 inhabitants due to gang presence. These results show that the local governments' capacity to spend on essential services and infrastructure is indeed constricted in the presence of gangs as their ability to collect taxes and generate income is hampered by gang-imposed toll taxes. Finally, we also found a significant negative effect on local economic activity and spillover effects to neighboring localities. Results indicate that for every homicide that occurs in one locality attributed to MS-13 and to gangs in general, the consumption of electricity within the local territory is significantly reduced. Considering that the average consumption per municipality in a year is 29 million kWh, this reduction represents a reduction of almost 10% in local activity. Gang activities had a negative impact not just on the municipality in which they operated, but in adjacent and neighboring municipalities as well; the more complex the organization of the gang that controls the territory, the greater the negative spillover in a micro region.

Taken together, the results depart from recent literature that has emphasized the symbiotic and complementary nature of the relationship between gangs and the state (Blattman et al.,

2021, and Lessing, 2021) and instead portray a much more parasitic relationship. Gangs like MS-13 that do not derive significant revenues from drug trafficking engage in such extensive extortion that they end up hollowing out municipal finances. Our findings offer additional support for Di Cataldo & Mastrorocco (2021), which suggests that the presence of gangs sharply restricts and limits the power of local officials in the territory they control. The presence of a strong and credible informal enforcer sways citizens and firms to comply with the demands from gangs instead of complying with formal municipal levies and fees. As Bueno (2022) reports, the extent and regularity of extortion suggests that firms may actually be including this payment as part of the cost of doing business. Finally, our findings offer additional support for Amaya & D'Aubuisson (2022), who argue that local businesses and households which suffer from extortion tend to remain in their municipality given that they lack the financial resources to fund their move to another jurisdiction, and because there is always a risk of moving to a municipality that suffers gang activity. The inability to move implies that the negative impact of gangs' crime activity is magnified upon neighboring municipalities due to the level of uncertainty that the presence of gangs represent in a micro region.

5. Conclusions and implications

This paper provides empirical evidence that the fiscal health of municipal governments in El Salvador is negatively impacted when gangs control territory, extort households and businesses, and impede the movement of goods. Through multiple causal mechanisms, widespread extorsion by Salvadoran gangs has prevented decentralization from delivering on its theoretical promise, even as the size of the effects depends a great deal on the size of the municipality in question. For example, own-source revenues for large municipalities (population greater than 45,000 inhabitants) are less sensitive to gangs' criminal activities because they have greater economic activity and enjoy more administrative capacity to generate their own revenues. Given their greater territorial size, it is easier for large municipalities to coexist with neighborhoods controlled by gangs. Disparities observed among municipalities of different sizes necessitate further investigation into the intricate interplay between economic factors, administrative capacity, and territorial characteristics that influence municipal sensitivity to gang-related issues. Our findings suggest that the relationship between gang rule and state rule (Blattman et al., 2021) is likely subject to significant cross-municipal variation, which has yet to be adequately theorized but which should ideally inform more differentiated policy responses from the central state.

In practice, far from offering differentiated policy responses that would help municipalities meet the challenges they face in the context of gang violence, the response by the national government to the patterns documented in this article has instead been to recentralize fiscal authority. As president since 2019, Nayib Bukele has introduced sharp cuts in fiscal transfers (FODES) and forced municipalities to seek central government permission via the new Municipal Works Directorate to invest any local revenues in the promotion of local development. Fiscal and administrative recentralization has weakened the financial pillar of decentralization in El Salvador, namely intergovernmental transfers, and increased the discretionary powers of the national government as the best way to combat the gangs. In 2023, the number of municipalities was also unilaterally decreased from 282 to 44 municipalities.[10] Recentralization with militarized control of the territory, combined with other 'iron fist' policies such as the suspension of civil rights, arguably have brought relief and security to many municipalities that lived in fear of extortion and other crimes by gangs in their neighborhoods. Moving forward, however, the key question is whether a policy centered on fiscal recentralization, the disappearance of the post-civil war decentralization framework, and the introduction of harsh security and military measures is enough to resolve all of the challenges in Salvadoran municipalities.

Our analysis suggests that an alternative, and perhaps complimentary, option is not to abandon decentralization but to reconstruct it for greater success. Even as security remains the main

priority of the current administration, a series of policy reforms could be introduced to improve the decentralization framework, including the modernization of the municipal tax code through the progressive introduction of an urban property tax (which may be less subject to extortion), revising the formula for revenue sharing to aid the medium-size municipalities that are especially vulnerable to extortion, and insulating the Municipal Works Directorate from political pressure so that it uses technocratic criteria in responding to municipal investment proposals. We conclude that instead of giving up on decentralization, which played such an important role in transforming post-conflict El Salvador, a viable alternative is to refine the decentralization framework and to more fully exploit complementarities with the national government. Resolving the security challenges created by gangs in El Salvador presents the country with an opportunity for a new beginning in decentralization.

Notes

1. For helpful comments on this article, we would like to thank Axel Radics and Alejandro Rasteletti of the Fiscal and Municipal Management Division of the Inter-American Development Bank. This publication does not constitute an endorsement by the InterAmerican Development Bank of the opinions expressed in it. Data and code available upon request at huanqui_sr@up.edu.pe.
2. For a comprehensive review of the relationship between decentralization and development, see Faguet 2014.
3. For the argument that regional inequality in developing countries can also impede factor mobility, see Rodríguez-Pose and Ezcurra 2010.
4. The municipal code assigns a number of other key roles to municipalities, including trash collection, municipal policing, the construction of housing, the regulation of local transport, commercial activities and restaurants, and the promotion of industrial development, education, and health (Vargas, 2022).
5. In terms of local expenditure, the main functions of municipalities are: solid waste removal, public safety in local markets, parks and local streets, construction and maintenance of cemeteries, fostering local economic development, and supporting community housing services.
6. Panel data could not be used, given that the spatial autocorrelation models in any given period only function with a balanced panel, and the availability of data is limited for the variables considered.
7. Nighttime lighting was obtained from Google Earth Engine and the Administrative Map from OCHA Latin America and the Caribbean. This variable is constructed using satellite recognition of faint levels of brightness on the Earth's surface during the night, which reveals a city's level of nighttime luminosity. This variable differs from the measurement of energy consumption, which is an indicator of economic activity (which takes place mostly during the daytime). It is worth remembering, however, that both variables may be correlated to the existence of urban centers.
8. The effect of gangs in public service in other size municipalities is marginally significant with a level of confidence less than 10%.
9. It is important to consider that there are significant variations in electricity consumption among localitites. In aggregate terms, the average electricity consumption per department varies, with some departments reaching as high as 120 million kWh, such as the case of San Salvador (or 972 kWh per capita), while other departments like Morazan record a consumption of 2.5 million kWh (or 256 kWh per capita).
10. Decree 762. June 14, 2023. https://www.asamblea.gob.sv/sites/default/files/documents/decretos/4194112C-1F6E-4E24-808E-9854A3D081AD.pdf.

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Appendix

1. Sources of information

A database was constructed for 2010–2018 using diverse sources of information (subsequent years were not considered due to the possible distortions that may have been caused to different variables by the effects of the COVID pandemic). These include the following:

- Fiscal Transparency Portal (El Portal de Transparencia Fiscal), administered by the El Salvador Treasury Ministry (Ministerio de Hacienda) for fiscal information at the municipality level.
- The Salvadorian Institute of Municipal Development (ISDEM) (Instituto Salvadoreño de Desarrollo Municipal) for information regarding the annual allocation of the Economic and Social Development Fund (FODES).
- The National Police Open data (Data Abiertos de la Policía Nacional) for the total number of homicides committed by the Mara Salvatrucha, used as a proxy for the municipal presence of the gangs.

- The General Superintendence of Electricity and Telecommunications (SIGET) (Superintendencia General de Electricidad y Comunicaciones), whose annual bulletins provided the information regarding electricity consumption at the municipal level, used as a proxy variable for economic activity.
- Google Earth Engine, using the Google Earth Private platform for the information concerning local night luminosity.
- National Institute of Statistics and Census (Dirección General de Estadísticas y Censos) to obtain local population census data.
- The Multi-purpose Household Survey (EHPM) (Encuesta de Hogares de Propósitos Múltiples), conducted by the Ministry of Economy to obtain information regarding the socio-economic situation of Salvadorian households, which is representative of the entire country.

2. Definition of variables and information source

Information source	Variable	Description
Fiscal Transparency Portal (Portal de Transparencia Fiscal)	IngCorr	Current Revenues (excluding FODES) (USD)
	GastCons	Consumption or Operational Management Expenditures (USD)
	Gasto	Sum of Consumption Expenditures and Financial Expenditures
	Tax_rev	Tax Revenues (USD)
	NoTax_rev	Non-Tax Revenues (USD)
	Own_rev	Own Revenues, understood as the percentage of tax revenues to total revenues (tax and non-tax)
	Own_rev_pc	Per Capita Collection, understood as tax revenues per inhabitant
	Cap_Exp	Capital Expenditures (USD)
	Current_transfer	Current Transfers (USD)
	Capac_build	Investment in Human Capital (USD), a proxy for the municipality's capabilities
	PC_rev	Per Capita Income
The Salvadorian Institute of Municipal Development (ISDEM)	FODES	Annual allocation to FODES (USD)
The National Police Open data	Maras	Homicides committed by Mara Salvatrucha (individuals)
	Other_gangs	Total homicides committed by all gangs
The General Superintendence of Electricity and Telecommunications	Kw	Energy consumption (millions of Kwh)
Google Earth Engine	Luminosity	Luminosity
National Institute of Statistics and Census	Popul	Population of the local jurisdiction
The Multi-purpose Household Survey	Density	Inhabitants per km2
	IncomeH	Family Income
	PubService	Percentage of inhabitants who dispose of garbage through a public collector, a proxy for the provision of public services
	Urban_popul	Urban Population (individuals)
	Agedpopulation	Percentage of the population over 65 years old

Authors' own elaboration.

3. Discussion of endogeneity

In our study, we acknowledge the complexity inherent in isolating causal relationships within the dataset and have applied rigorous methods to mitigate the influence of endogenous factors. Here we include supplementary analyses that validate our conclusions and reinforce the consistency and robustness of our findings.

3.1. Effects of gangs on local finance: Own-source revenues

For the effects of gangs on local finance, additional estimations were conducted using two-stage least squares (2SLS) method to examine the impact of gang presence on per capita income. The use of instrumental variables allowed us to address potential endogeneity issues and strengthens the credibility of these results in shedding light on the complex interplay between gang presence and fiscal local outcomes. These findings suggest that the presence of the MS-13 gang has a discernible and adverse impact on the level of personal income, while the overall effect of all gangs combined on per capita income is present but not statistically robust.

2SLS for the effect of gang violence on per capita own revenue

	MS-13	All gangs
ln_gangs	−0.609*	−0.231
	(0.270)	(0.290)
ln_urban	0.159	0.031
	(0.085)	(0.117)
ln_incomeh	0.437*	−0.335
	(0.181)	(0.338)
ln_kw	0.655***	0.914***
	(0.090)	(0.206)
ln_capac_build	−0.037	−0.114
	(0.033)	(0.087)
ln_transfers	0.735***	0.648
	(0.126)	(0.368)
ln_density	−0.035	0.070
	(0.040)	(0.076)
ln_pubservice	0.057	0.208
	(0.069)	(0.140)
Constant	−5.234***	0.709
	(1.487)	(3.024)
Observations	446	44

Standard errors in parentheses. Robust standard errors in parentheses.
$*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

3.2. Effects of gangs on local finance: Local expenditure

An IV estimation was conducted to account for the potential endogeneity issues between homicides and per capita expenditure. Similarly, to the revenue model, a two-stage least squares (2SLS) estimation with IV was applied, considering gang homicides (both Maras and all gangs) with a lag of two years in each case. The results demonstrate relative consistency, indicating that only the model using data for all gangs shows significant and negative effects on local per capita expenditure. However, no significant relationship was found between homicides attributed specifically to the Maras and per capita expenditure.

These findings suggest that gang-related homicides, when analyzed in conjunction with all gangs, have a discernible effect on reducing local per capita expenditure. However, the specific impact of Maras-related homicides on expenditure remains inconclusive.

Effect of the gang presence on local expenditures – IV

	MS-13	All gangs
ln_gangs	−0.240	−0.137*
	(0.135)	(0.062)
ln_urban	0.000	0.028
	(0.041)	(0.040)
ln_kw	0.246***	0.242***
	(0.049)	(0.039)
ln_capac_build	−0.022	−0.022
	(0.014)	(0.015)
ln_transfers	0.693***	0.754***
	(0.071)	(0.077)
ln_density	−0.005	−0.007
	(0.021)	(0.017)
Constant	1.306*	0.748
	(0.580)	(0.622)
Observations	451	387

Standard errors in parentheses. Robust standard errors in parentheses.
*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.